

Scalability & Future Plan

Date	03 October 2026
Team ID	FITBUDDY-TEAM
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	1 Marks

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	SQLite is suitable for the current local application but not ideal for large concurrent workloads.	Database scaling would become a bottleneck.	High
2	AI response quality and latency depend on the Gemini service.	External service availability affects generation.	High
3	Current fitness input is relatively limited.	Plans can be improved with richer user preferences and progress data.	Medium
4	The supplied project is demonstrated locally rather than as a production cloud service.	Public multi-user access requires deployment hardening.	Medium

Future Enhancements

- PostgreSQL for production-scale persistence.
- Cloud deployment with monitored API services.
- Progress tracking, workout completion history and charts.
- More detailed exercise constraints and equipment preferences.
- Goal progress feedback over multiple weeks.
- Optional wearable/device integration.
- Improved authentication, role-based coach access and audit logging.